

Titanic Data Cleaning & Passenger Analysis

Objective

To clean the Titanic dataset and analyze passenger survival trends based on factors such as age, gender, and passenger class.

Dataset

Titanic Dataset obtained from Kaggle.

Data Cleaning

- Loaded the dataset using Pandas.
- Checked for missing values.
- Filled missing values in Age and Embarked columns.
- Removed the Cabin column due to excessive missing values.
- Checked and removed duplicate records.
- Identified and treated outliers using boxplots and the IQR method.

Statistical Analysis

- Calculated descriptive statistics such as mean, median, minimum, and maximum values.
- Analyzed the distribution of passenger ages and fares.
- Calculated overall survival rates.

Grouping Analysis

- Grouped passengers based on gender and passenger class.
- Compared survival rates across different groups.
- Analyzed average age within each passenger class.

Visualizations Created

- Survival Count Plot
- Gender Distribution Plot
- Passenger Class Distribution Plot
- Age Histogram
- Age Boxplot

Key Observations

1. Female passengers had a significantly higher survival rate than male passengers.
2. First-class passengers were more likely to survive than second- and third-class passengers.
3. Younger passengers generally showed better survival chances.
4. Most passengers belonged to the young and middle-aged groups.
5. Passenger class and gender were the most important factors influencing survival.

Conclusion

The analysis revealed that survival on the Titanic was strongly influenced by gender and passenger class. Female passengers and first-class travelers had the highest chances of survival. The project demonstrated the importance of data cleaning, statistical analysis, and visualization in extracting meaningful insights from real-world datasets.